



# Assignment 1

## Objective:

- Use a keypad to control LED patterns and display the current pattern name on a 16x2 character LCD.

## Components:

- ATmega32 microcontroller
- Keypad
- LEDs
- Character LCD 2 \* 16 display
- Resistors
- wires

## Circuit Setup:

- Connect the keypad to the ATmega32 for input.
- Connect multiple LEDs to output ports on the ATmega32.
- Connect the 16x2 LCD to the microcontroller for display.

## Programming:

- Initialize the keypad for user input.
- Initialize the LEDs and LCD for output.
- Map each keypad button to a specific LED pattern.
- Implement functions for each LED pattern (e.g., blinking, running light).
- Display the name of the active pattern on the LCD.

## Pattern Control :

- Use keypad input to switch between different LED patterns.
- Ensure the LCD updates to show the name of the selected pattern.